



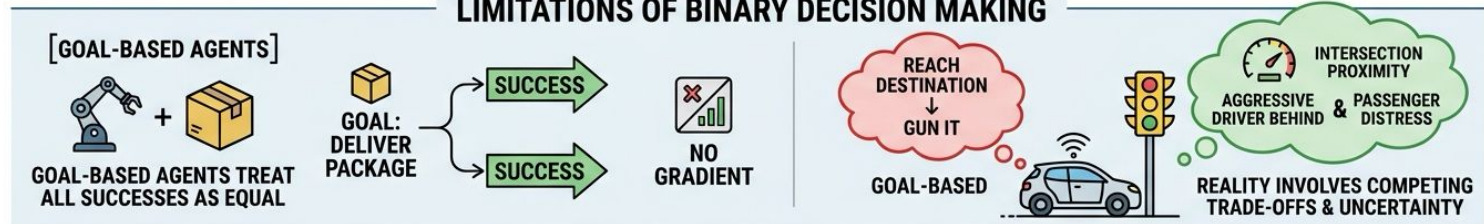
U.S. General Services
Administration

Chapter 5.10: Utility-Based Decision Making Fundamentals

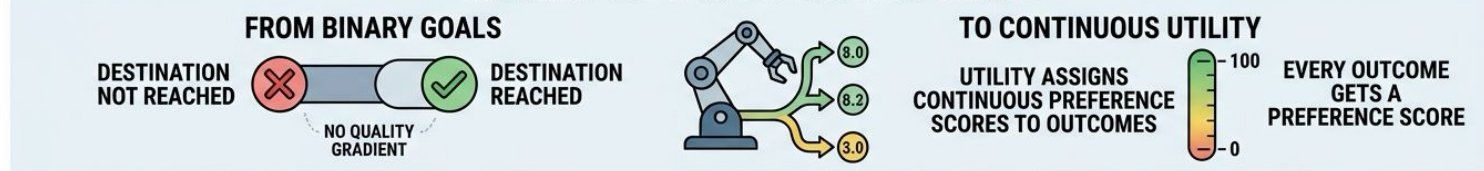
US General Service Administration
AI Community of Practice - March 2026

From Binary Goals to Continuous Utility

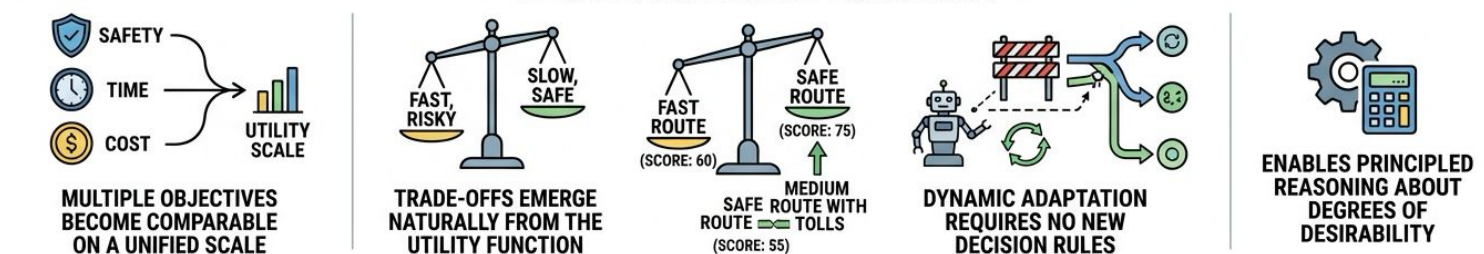
LIMITATIONS OF BINARY DECISION MAKING



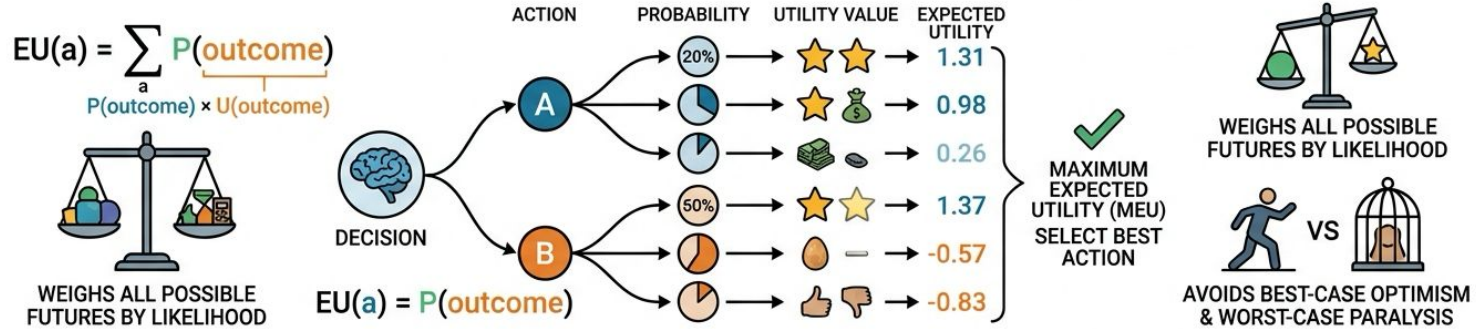
TRANSITIONING TO CONTINUOUS UTILITY



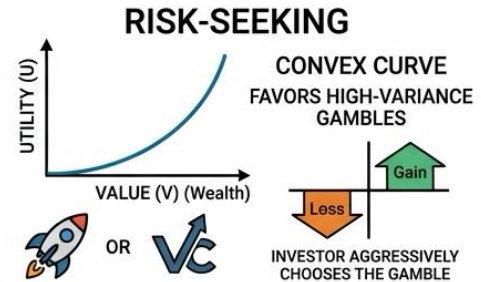
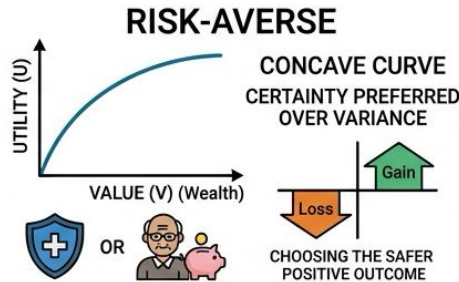
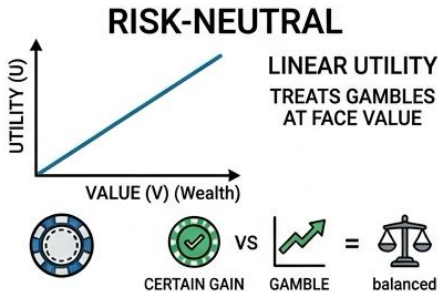
BENEFITS OF UTILITY FUNCTIONS



Expected Utility -- Probability Meets Preference



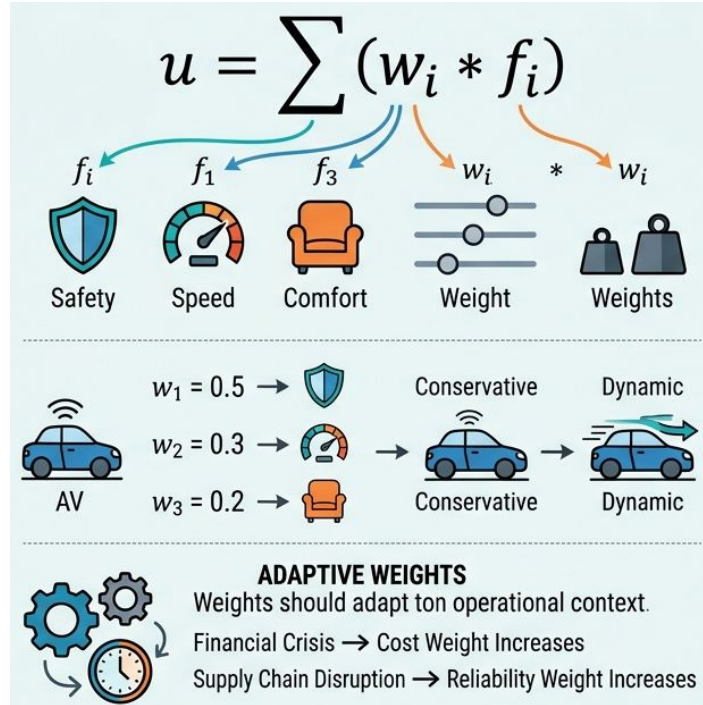
RISK ATTITUDES SHAPE UTILITY CURVE



CURVE SHAPE MUST MATCH THE PROBLEM DOMAIN



Multi-Objective Utility -- Weighted Trade-offs



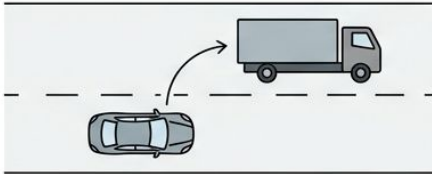
- Combine objectives: $u = w_1 * f_1 + w_2 * f_2 + \dots + w_k * f_k$
- Weights encode trade-offs directly between objectives
- Adjusting weights changes agent behavior systematically
- Example: safety 0.5, speed 0.3, comfort 0.2 for autonomous vehicles
- Weights should adapt to changing operational contexts

Common Pitfalls That Wreck Utility Systems

- Confusing monetary value with subjective utility
 - Defaulting to linear utility when the domain demands curvature
 - Treating probability estimates as exact known quantities
 - Ignoring that Markov assumptions may not hold
 - Skipping validation against historical decisions
- 

Case: Autonomous Vehicle Decisions

LANE CHANGE



Large efficiency gains from small safety trade-off.

EU = 0.82



0.8

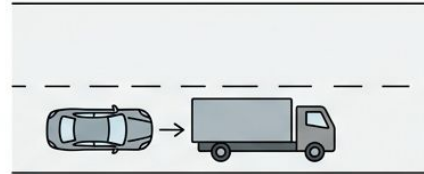


0.95



0.9

STAY IN LANE



Large efficiency gains from small safety trade-off.

EU = 0.65



1.0

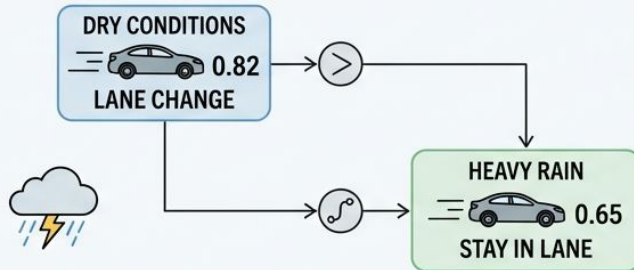


0.3

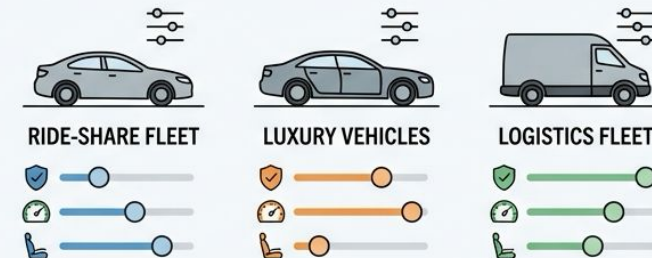


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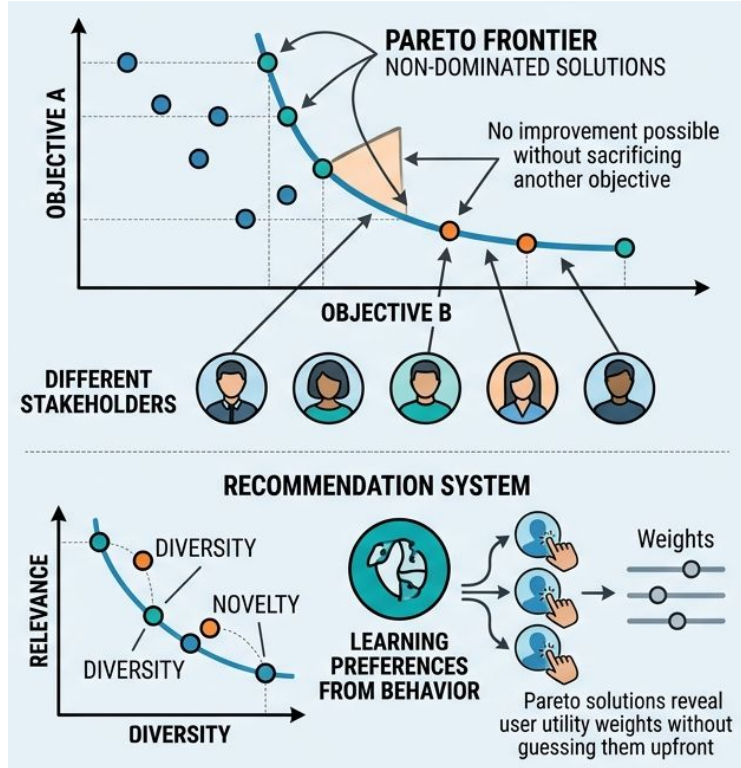
RECALCULATES AUTOMATICALLY WHEN CONDITIONS CHANGE



FLEET OPERATORS TUNE WEIGHTS FOR DIFFERENT VEHICLE ROLES



Beyond Weighted Sums -- Pareto Frontiers



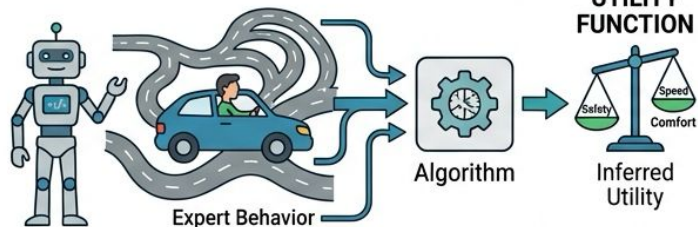
- Weighted sums hide value judgments in weight selection
- Pareto optimal: no improvement without sacrificing another objective
- The Pareto frontier shows all non-dominated solutions
- Different stakeholders can pick different points on the frontier
- Algorithms like NSGA-II compute frontiers for large problems

Learning the Utility Function Itself



INVERSE REINFORCEMENT LEARNING

OBSERVE & INFER

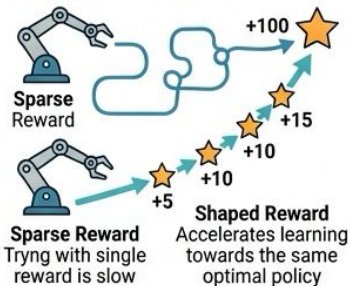


MULTIPLE EXPLANATIONS



ACCELERATING LEARNING & ADAPTATION

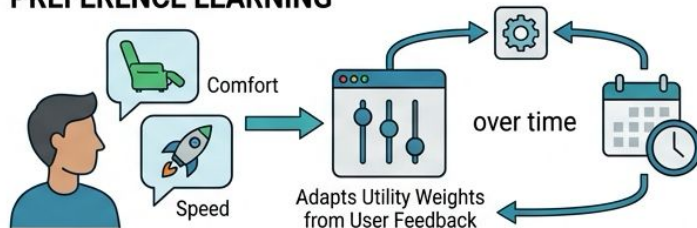
REWARD SHAPING



POTENTIAL-BASED SHAPING



PREFERENCE LEARNING



When Utility Beats Rules -- and When It Does Not

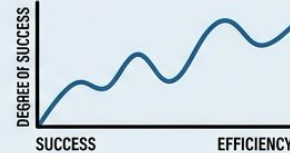
COMPLEX TRADE-OFFS



UNCERTAIN OUTCOMES



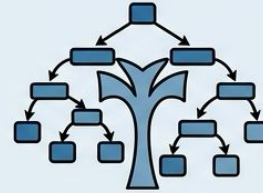
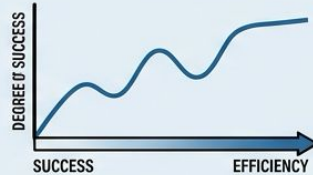
CONTINUOUS OPTIMIZATION



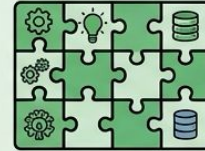
BINARY DECISIONS



UNCERTAIN OUTCOMES



COMPLETE DOMAIN KNOWLEDGE

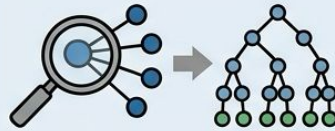


CAREFUL CALIBRATION & VALIDATION



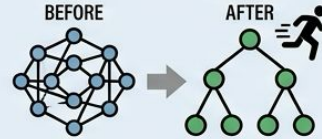
UTILITY FUNCTIONS REQUIRE CAREFUL CALIBRATION AND VALIDATION

COMPUTATIONAL COST GROWS



COMPUTATIONAL COST GROWS WITH ACTION SPACE AND OUTCOME ENUMERATION

APPROXIMATION METHODS



APPROXIMATION METHODS TRADE SOLUTION QUALITY FOR REAL-TIME

TRANSPARENCY CRITICAL



Key Takeaways

- Utility functions turn binary goals into continuous optimization
 - MEU selects actions by probability-weighted desirability
 - Risk attitude is encoded in utility curve shape, not ad-hoc rules
 - Multi-objective trade-offs need explicit weights or Pareto analysis
 - Next chapter: rule-based decision making offers complementary strengths
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